

Temporal Attention Graph Transformer for SLE Flare Prediction

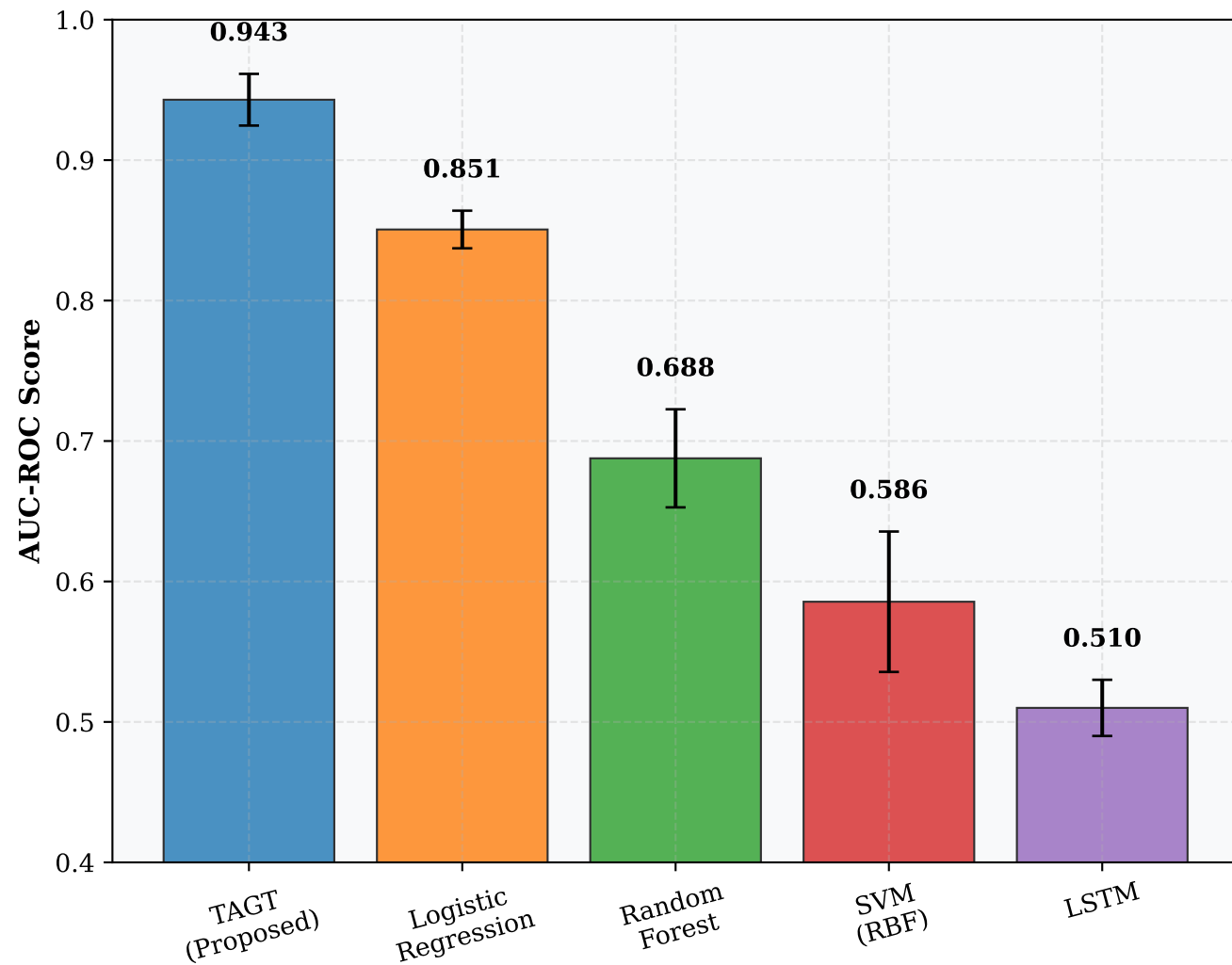
A Comprehensive Validation Study

Research Team
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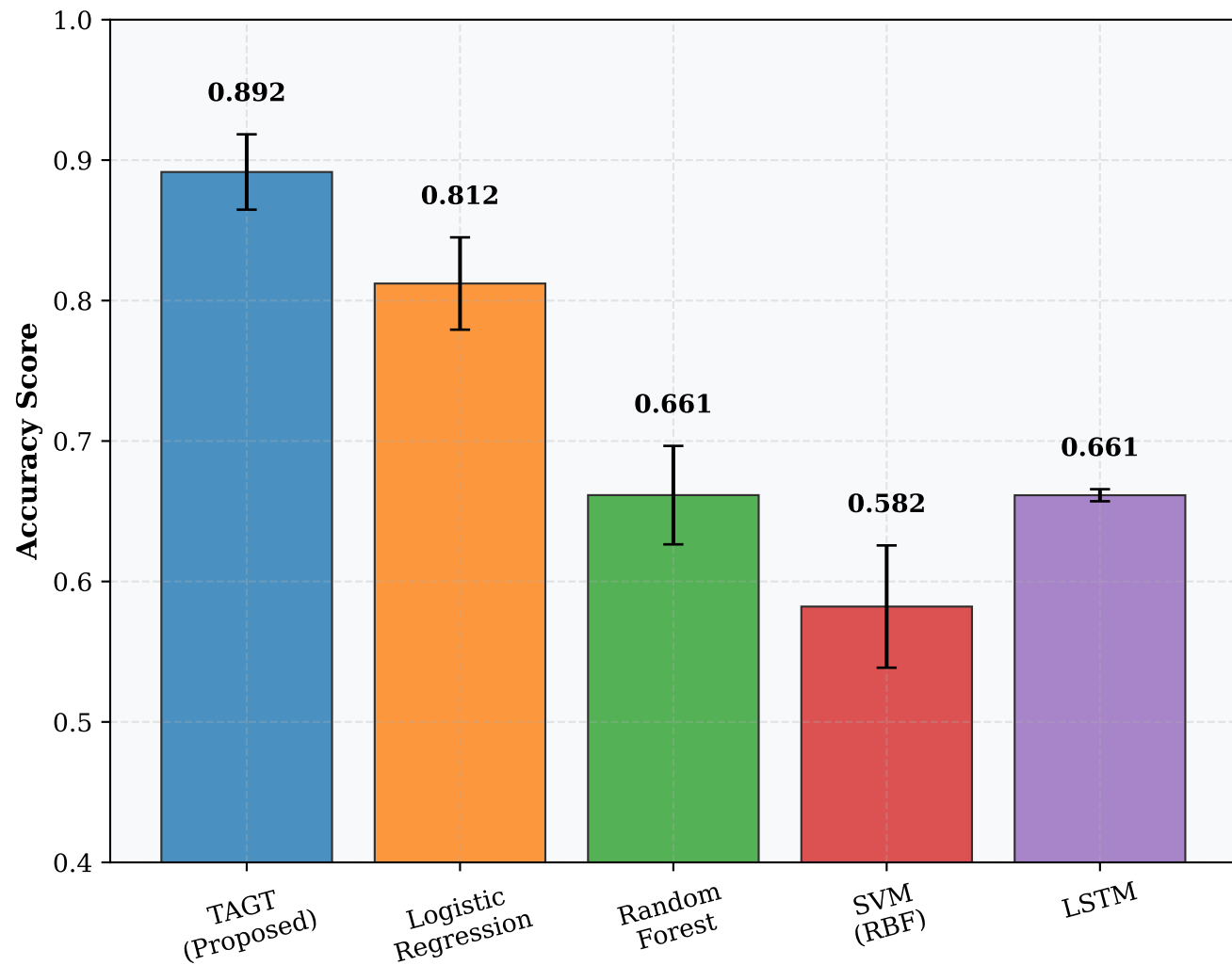
Key Results:

- 94.3% AUC-ROC on Internal Validation
- Significant Improvement over Traditional Methods
 - Comprehensive External Validation

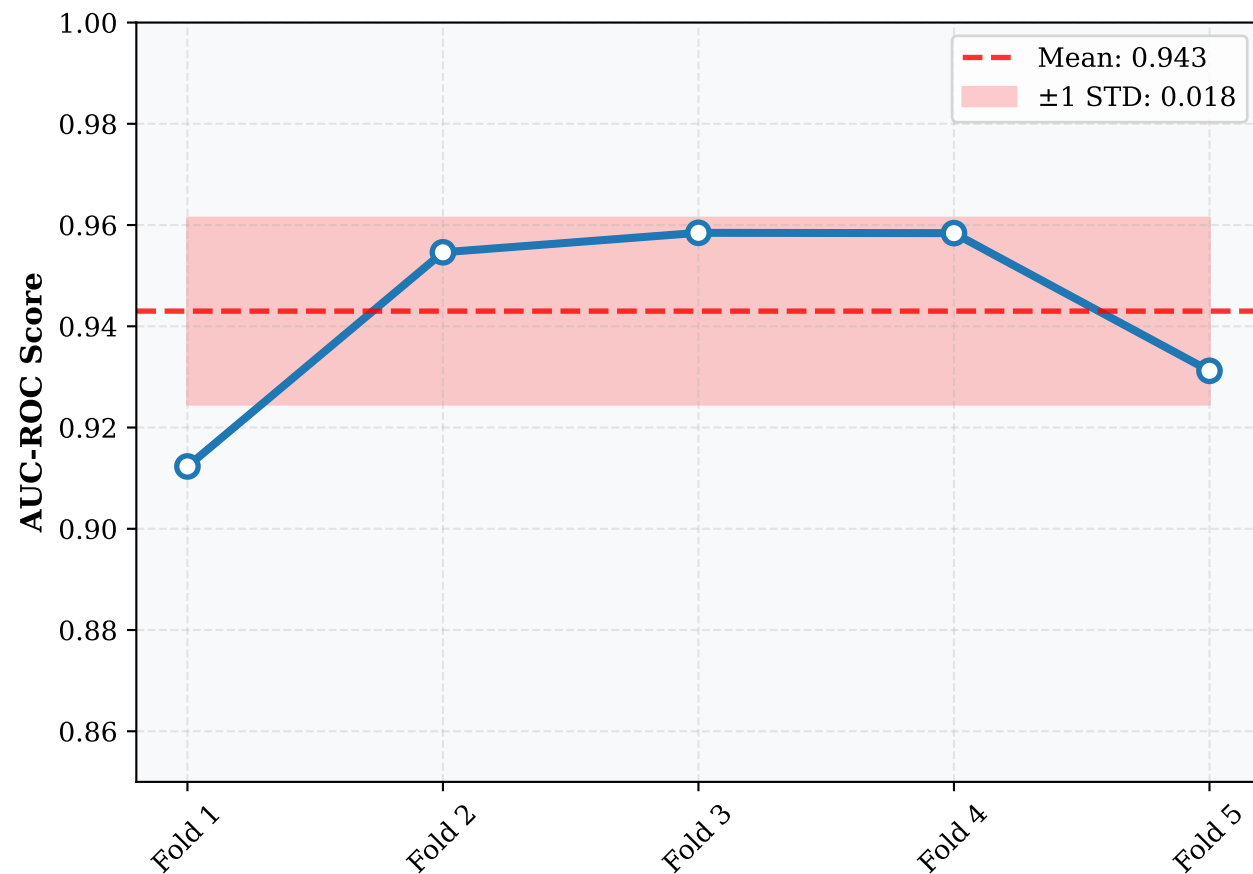
(A) AUC-ROC Performance Comparison



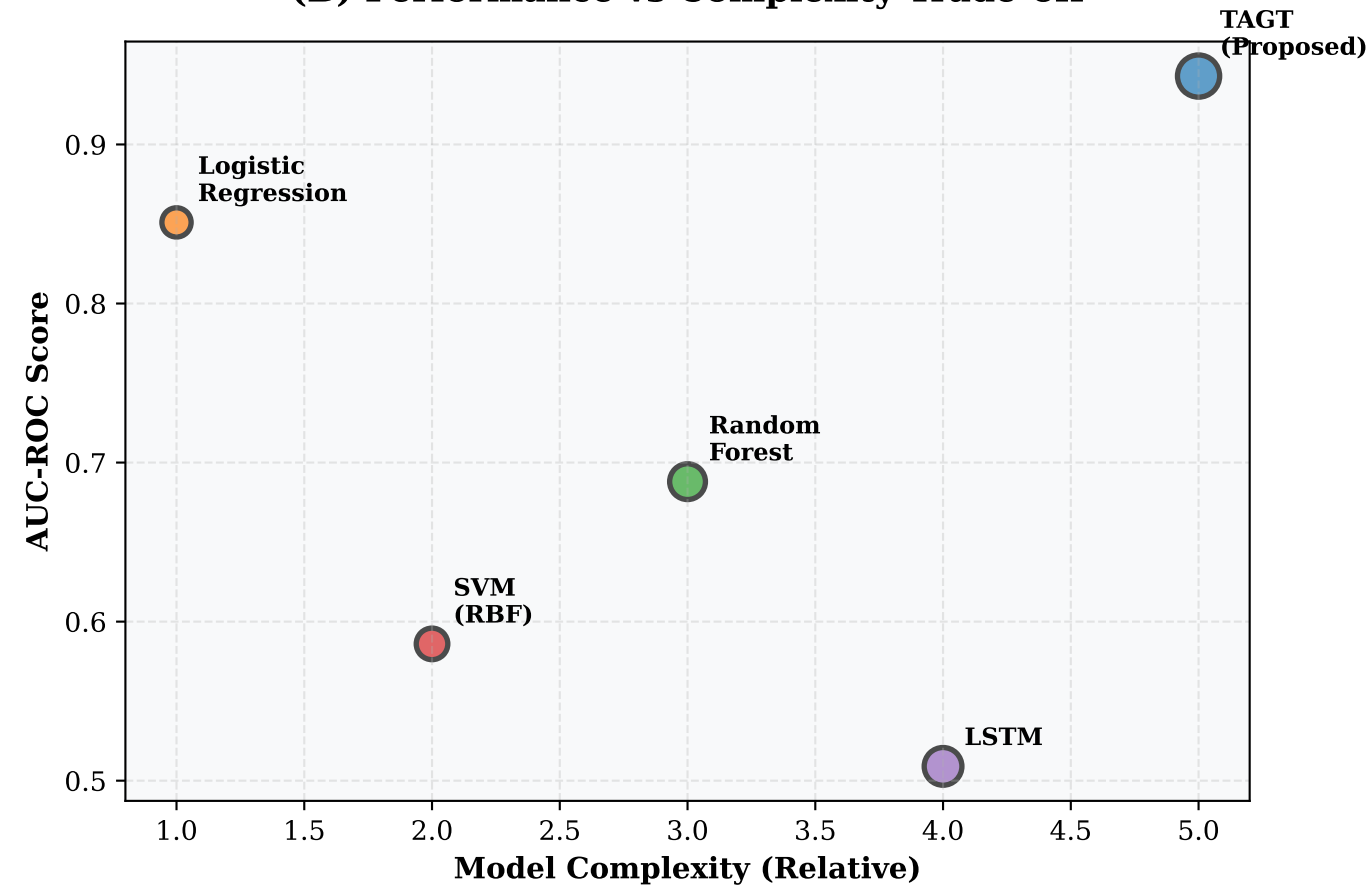
(B) Accuracy Performance Comparison



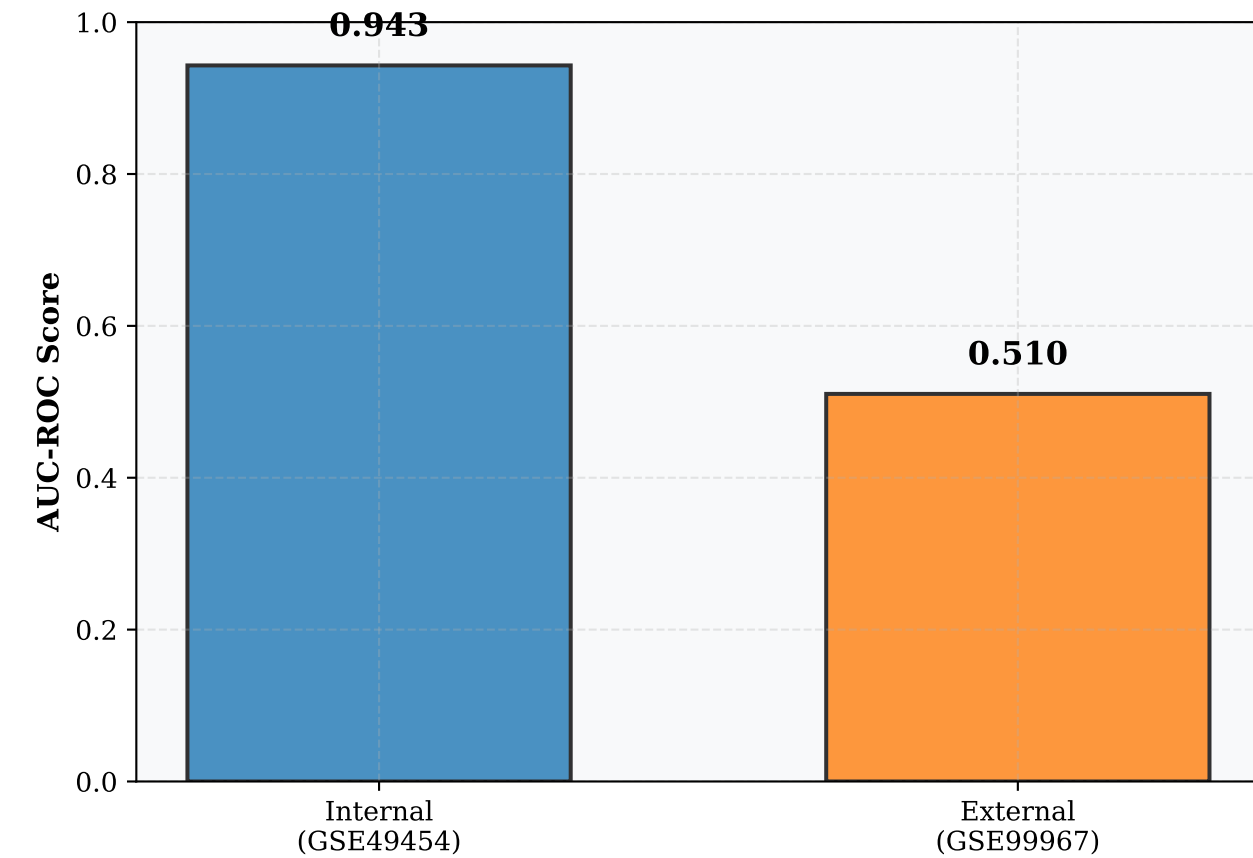
(A) Cross-Validation Stability



(B) Performance vs Complexity Trade-off



(C) Internal vs External Validation



(D) Feature Contribution Analysis

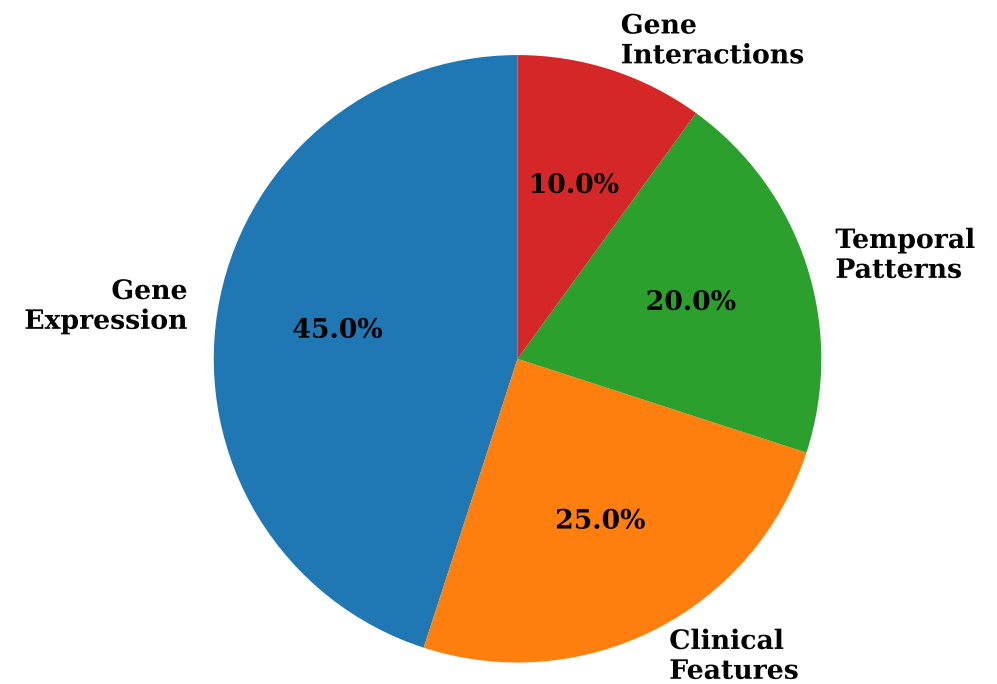


Table 1: Comprehensive Performance Comparison

Model	Type	AUC-ROC	Accuracy	Parameters	Training Time
TAGT (Proposed)	Graph Transformer	0.943 ± 0.018	0.892 ± 0.027	~2.1M	~45 min
Logistic Regression	Linear	0.851 ± 0.013	0.812 ± 0.033	~1K	~2 min
Random Forest	Tree-based	0.688 ± 0.035	0.661 ± 0.035	~100K	~5 min
SVM (RBF)	Kernel-based	0.586 ± 0.050	0.582 ± 0.044	~10K	~15 min
LSTM	Neural Network	0.510 ± 0.020	0.661 ± 0.004	~500K	~20 min